



Sai Nikhil Mattapalli

AI/ML ENGINEER

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PROFESSIONAL SUMMARY

AI/ML Engineer and Data Scientist with 5+ years of experience developing machine learning models, production AI applications, and data-driven solutions. Experienced in building and deploying production LLM applications, Retrieval-Augmented Generation (RAG) pipelines, agentic AI systems, and predictive analytics models using Python, SQL, TensorFlow, Scikit-learn, LangChain, Azure OpenAI, AWS, and Azure. Skilled in end-to-end model lifecycle management, MLOps, model deployment, optimization, and AI platform development to support decision-making and operational efficiency.

SKILLS

Methodologies:	SDLC, Agile, Waterfall
Programming Language:	Python, SQL, R
Packages:	NumPy, Pandas, Matplotlib, SciPy, Scikit-learn, TensorFlow, Seaborn, ggplot2, Plotly, Keras, OpenCV, NLTK
ML/Generative AI:	Regression analysis, Bayesian Method, Decision Tree, Random Forests, Support Vector Machine, Neural Network, Sentiment Analysis, K-Means Clustering, KNN, Natural Language Processing (NLP), LangChain, OpenAI API, Hugging Face Transformers, Prompt Engineering, RAG Architecture, LLM Workflows, Semantic Search, Embeddings, Vector Databases (FAISS, Pinecone), LoRA, PEFT, instruction tuning
Visualization Tools:	Tableau, Power BI, Advanced Excel (Pivot Tables, VLOOKUP)
IDEs:	Visual Studio Code, PyCharm, Jupyter Notebook, IntelliJ
Cloud Technologies:	Amazon Web Services (AWS S3, EMR, Redshift), Microsoft Azure
Database:	MySQL, Oracle DB, PostgreSQL, MongoDB
Other Technical Skills:	Predictive Analysis, Time Series Forecasting, Regression Analysis, Hypothesis Testing, Classification, Clustering, Principal Component Analysis (PCA), Deep Learning, Model training and evaluation, SSIS, SSRS, Hadoop, Spark, Kafka, Snowflake, Big Query, Apache Airflow, Informatica, Talend, Jenkins, Kubernetes, Docker, Git

WORK EXPERIENCE

AI/ML Engineer | Goldman Sachs, USA

Aug 2023 - Present

- Developed an AI-powered Financial Research & Risk Intelligence Platform using Python, LangChain, Azure OpenAI, and Hugging Face Transformers to automate financial knowledge discovery and analysis.
- Built Retrieval-Augmented Generation (RAG) pipelines using LangChain, OpenAI embeddings, and vector databases, achieving 35% higher information retrieval accuracy for financial research and knowledge discovery applications.
- Designed scalable ETL pipelines using PySpark, Python, SQL, and Apache Spark to process over 2 TB of structured and unstructured financial data.
- Leveraged JAX for accelerated model training and large-scale financial data processing, reducing experimentation and training time by 30%.
- Constructed machine learning models for risk prediction, anomaly detection, and document classification using Scikit-learn and TensorFlow, improving prediction accuracy by 25% for financial risk assessment and fraud detection workflows.
- Implemented serverless orchestration using AWS Lambda and AWS Step Functions to streamline AI model inference and financial risk analysis workflows.
- Built multi-step reasoning workflows leveraging LLM agents and tool-calling capabilities to automate financial KPI analysis and investment research tasks.
- Fine-tuned Hugging Face Transformer models on documents and intelligence datasets, improving classification and information extraction accuracy by 25%.
- Deployed ML and Generative AI applications on AWS using Docker, MLflow, and CI/CD pipelines for scalable production environments.
- Utilized Vertex AI Pipelines for automated model training, deployment, and monitoring, accelerating model lifecycle management by 35%.

Data Scientist | Cognizant, India

Feb 2020 - Aug 2022

- Developed and deployed a real-time machine learning model using Python, TensorFlow, and scikit-learn to analyze market fluctuations based on historical price trends, order book dynamics, and macroeconomic indicators. Improved predictive accuracy by 15%, enabling traders to manage risk exposure and optimize hedging strategies.
- Engineered RL-driven execution model using OpenAI Gym and TensorFlow, dynamically adjusting trade placements based on liquidity conditions and market microstructure. Reduced transaction costs by 5%, increasing profitability for algorithmic trading strategies.
- Designed anomaly detection framework with Apache Kafka and PySpark, analyzing trade activity in real time to identify market manipulation and suspicious transactions. Reduced false positives by 30%, ensuring compliance with SEC & FINRA regulatory standards.
- Optimized SQL queries on a 100TB+ Azure Synapse Analytics data warehouse, restructuring partitioning, distribution, and indexing for improved parallel execution. Reduced query response times by 40%, accelerating data retrieval for risk management and regulatory reporting.
- Led statistical experiments using Python (NumPy, SciPy, StatsModels) to assess the impact of new trading signals on execution strategies. Improved market-making and arbitrage efficiency by 10%, enhancing overall trade performance.
- Created interactive Tableau dashboards integrating live trading metrics, risk exposure analysis, and PnL visualizations. Reduced stakeholder decision-making time by 25%, improving transparency in portfolio management and risk mitigation.
- Designed a Natural Language Processing (NLP) pipeline using BERT, NLTK, and financial news data to extract insights from earnings reports, market news, and social media sentiment. Increased sentiment prediction accuracy by 18%, aiding risk assessment teams in anticipating market shifts.

EDUCATION

Master of Science in Computer Science – Suny Albany at Albany, USA

Bachelor of Technology in Computer Science – Bharath University, India

CERTIFICATIONS

Google Professional Machine Learning Engineer Azure AI Engineer Associate (AI-300)